

PRATIKSHIT SINGH

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[LinkedIn](#) | [Github](#)

EDUCATION

- University of Illinois Urbana-Champaign, IL, United States** Aug 2022 - Dec 2023
Masters - Computer Science **GPA:3.9/4.0**
Courses: DBMS, Web Programming, Software Engineering & Testing
Generative AI, Computer Vision, Data Mining, Transfer Learning, AI
- Indian Institute of Technology (BHU) Varanasi, India** Jul 2016 - May 2020
Bachelor of Technology - Electrical Engineering **GPA:9.5/10.0**
Courses: NLP, Probability and Statistics, Number Theory, Control Systems, Numerical techniques

TECHNICAL SKILLS

- Programming Languages:** Python, Java, TS, JS, C, C++, SQL, GraphQL, Shell, Design Patterns(LLD, HLD)
- Technologies:** Springboot, Django, Flask, Hadoop, EMR, EC2, Lightsail, PySpark, Hive, Kafka, Redis, React, Jenkins, Docker, Kubernetes, Nginx, JUnit, Mockito, Karate
- genAI:** LangChain, LangGraph, Agents, RAG, Embedding, VectorDB, Evaluation, MCP, Harness Engineering
- Databases:** Design Patterns, Indexing, Snowflake, Postgres, MySQL, DynamoDB, MongoDB, S3, Redis, HMS, Datalake
- Libraries:** PyTorch, TensorFlow, Pandas, Numpy, Sklearn, NLTK

WORK EXPERIENCE

Intuit | Software Engineer Sep 2024 - Current
Privacy Engineering

- Developed Claude Plugins for Legal, Privacy Counsels, and Product teams to accelerate manual reviews, triaging users queries in a self-serving way and building a precise & greater velocity oriented organization.
- Spearheaded the **design & development** of **Intuit Complaint Management Platform** processing **100GBs of transcript data/day** from **datalake** with **PySpark EMR** pipelines with **LLM models, RAG** to generate insights, & regulatory checks.
- Implemented **RAG-powered** Legislative bill tracking to identify upcoming law changes for privacy, accelerating legal compliance review process by **60%**.
- Spearheaded the **design & development** of multiple **event-driven, MVC** and **Hexagonal architected** SpringBoot microservices with **99.99% availability**, multi-region setup, throttling, rate-limiting API calls.

Zeta Global | Software Engineer June 2024 - Sep 2024
Zeta Marketing Platform

- Designed and implemented **python ETL EMR pipeline** to identify brands & category statistics in email campaigns, utilizing OCR and openAI apis to **enrich BI reporting for customers..**
- Tech Stack:** Apache Superset, Flask, OCR, DynamoDB, Boto3, openAI

Illinois Center for Transportation, UIUC | Lead Developer Dec 2022 - May 2024
Pavement Life-Cycle Assessment Tool [[LCA application](#), [IPR application](#)]

- Led the design & development of scalable multi-stage complex web applications for Pavement Life & Cost Assessment.
- Used DRF to process requests & fetch data from Postgres, with Redux store for managing hierarchical project data in UI.
- Implemented user permissions for calculation models & tool stages with caching deployed as a multi-container application.
- Tech Stack:** Django, REST API, PostgreSQL, Redis, React, Redux, AWS LightSail, Docker Compose

JP Morgan Chase & Co | Software Engineer Aug 2020 - Jul 2022
Quantitative Tool for Risk and PnL - Full Stack Developer

- Led the development of Athena QIT web application, created trades dashboards to view trading risk & market exposures for 15+ Global Trading Desks, using async tornado web framework and Athena Risk Engines for adhoc risk analysis.
 - Designed & implemented Postgres database table & views using feeds from 200+ risk jobs. Further, implementing DB triggers for updating tables from live running risk jobs.
 - Accelerated data delivery to dashboards using clustered indexing for SQL and GraphQL API calls.
 - Savings** - Annual cost of \$5 Million by decommissioning legacy third-party applications.
 - Stack:** Tornado-web-server, React, PostgreSQL, GraphQL, DataDog, SDLC, Automated Testing
- Athena Risk Compute Batches Optimization using ML clustering algorithm - Data Scientist
- Built an EOD cron job for optimizing Risk Jobs for IR instruments using XGBoost and Linear regression models, to generate optimized batches to be processed on EC2 pods.
 - Savings** - 84% reduction in total running time for Global jobs running on the Amazon EC2.
 - Stack:** Feature Creation, Regression Models, Elastic-Net, AWS EC2, Cloud Computing, Kubernetes

PROJECTS

Attention Enhanced-Super Resolution GAN [Github](#)

- Implemented SRGAN with multi-scale attention U-net discriminator & residual-in-residual dense blocks generator.
- Better- NIQE score over current SR models & accurately generated sharp edge details & textures.
- Tech Stack:** Super Resolution, GAN, Attention blocks, RRDB, U-Net, CNN

Domain Adversarial Transfer Learning for Time Series Classification [Github](#)

- Developed methods to generate superior pre-trained 1-D CNN transfer models for time-series classification.

- **Tech Stack:** Domain adversarial networks, Error bounds, Timeseries, DTW & divergence measures, CNN

Fraud Detection on Imbalanced Class Dataset

[Github](#)

- Implemented SMOTE sampling, cross-validation methods, and cost-sensitive models on big data with 0.02% positive class.
- **Accuracy:** 97.7 % Train and 95.1% Test with cost-sensitive XGBoost.
- **Tech Stack:** EDA, Pattern mining, SMOTE, Cost-sensitive XGBoost, PCA

Deep Learning-based Fault Detection in Transmission Towers

Power Grid Corporation of India Ltd.

- Implemented CNN CV model using TensorFlow and YOLOv3 network to detect faults in various classes of Transmission Towers using high-resolution images.
- **Tech Stack:** Tensorflow, Keras, YOLOv3, CNN, sliding window, anchor box, IoU